

Stanislav Smirnov

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SUMMARY

Senior Data Scientist in payments and fraud with experience leading 0-to-1 ML and optimization products across routing, retries, decline recovery, tokenization and fraud detection. I combine ML and analytics with KPI ownership, stakeholder alignment, and production delivery to improve authorization, recover revenue, and reduce fraud.

HIGHLIGHTS

- Delivered a **4x** uplift in fallback routing success and **3x** increase in recovered revenue by analyzing PSP performance on NA traffic and redesigning a low-performing rerouting path.
- Originated and scaled the **Dynamic Retries** ML solution from idea to production, improving in-scope retry success by **25–50%** through KPI-led experimentation and rollout.
- Rebuilt a production fraud ML model improving fraud recall from **~44% to ~72%** at the same **1% false-positive rate** and increasing PR-AUC from **0.51 to 0.80**.
- Drove cross-functional delivery with Product, Engineering, and Fraud Prevention from prioritization and KPI definition through rollout and evaluation.

SKILLS

Leadership & Delivery: stakeholder management, cross-functional leadership, execution ownership, structured problem framing, mentoring junior data scientists

Product & Strategy: KPI definition, prioritization, decision memos / PRDs, rollout planning, launch evaluation, roadmap/backlog definition, experimentation design

Payments & Fraud: authorization/decline analysis, PSP performance, routing, retries, tokenization (network tokens), payment optimization, fraud detection

Programming & ML/AI: Python, SQL, PySpark, classical ML, Deep Learning (CNNs, LSTMs), LLMs

Cloud & Data: AWS, Azure, Data Vault 2.0, ETL development

Tools: Git, Docker, Airflow, FastAPI, Databricks, TensorFlow, Keras

EXPERIENCE

Senior Data Scientist, Payments

April 2024 – Present

Cleverbridge AG

Cologne, DE

- Lead payment optimization and fraud ML initiatives across routing, retries, authorization, tokenization, and fraud detection, owning work from opportunity discovery and decision memo through stakeholder alignment, rollout, KPI definition and evaluation.
- Improved fallback re-routing authorization success by **4x** (from 2–3% to 11%) by analyzing PSP performance on NA traffic and proposing a switch from a low-performing route to a higher-performing alternative; resulted in a **3x** increase in recovered revenue on that traffic slice.
- Initiated and led the rebuild of a production fraud ML model that had not been retrained for two years; worked with Fraud Prevention to redesign the fraud-labelling logic, retrained and deployed the model, and improved offline fraud recall from **~44% to ~72%** at the same **1% false-positive rate** (PR-AUC: **0.51 to 0.80**). The evaluation benchmarked the existing model and an external vendor score, supporting a build-vs-buy decision.
- Own the evolution of **Dynamic Retries** and use BIN-level, regional, PSP, and network-token performance analysis to prioritize additional routing and retry use cases.
- Lead and mentor junior data scientists through problem framing, model development, experimentation design, and production analytics; raised quality via code reviews and structured feedback.

Data Scientist

December 2021 – April 2024

Cleverbridge AG

Cologne, DE

- Originated and shipped **Dynamic Retries**, an ML-driven retry timing product; wrote the product proposal, aligned engineering, product, and business stakeholders, and led delivery from PoC through MVP to production rollout.
- Improved success rates for in-scope retries by **25–50%**, owning KPI definition, measurement design, and iteration based on post-launch performance.

- Designed a hybrid retry strategy combining a rule-based scheduling heuristic (country/currency × day-of-month × hour-of-day) with ML timing optimization to select high-conversion retry windows.
- Implemented and operated the solution on AWS (Python microservices, SageMaker, MWAA), including training/inference workflows and automation for reliable production runs.
- Designed the ML framework behind **CleverAutomations** and delivered churn-reduction models and automated workflows that identified at-risk customers and triggered targeted campaigns.

Data Engineer

September 2019 – December 2021

Cleverbridge AG

Cologne, DE

- Led the design and implementation of a scalable data warehouse on Azure (Data Vault 2.0), enabling reliable reporting and self-service analytics across the company.
- Built and optimized ETL pipelines (Azure Data Factory, Databricks, T-SQL) processing millions of records per day from multiple source systems.
- Managed schema changes and migrations with Flyway, ensuring consistent, versioned deployments across environments.
- Developed Python microservices and Airflow workflows for data products, containerized with Docker for resilient, maintainable operation.

Machine Learning Intern

May 2017 – July 2017

KOSTAL Group

Dortmund, DE

- Trained a gesture-recognition neural network using Keras and TensorFlow; achieved **80%** validation accuracy on top-5 gestures using static image inputs with a CNN.
- Improved model robustness through data preprocessing and augmentation.

Network Optimization Engineer

May 2012 – March 2015

MegaFon

Republic of Bashkortostan, Russia

- Led the **network optimization team** of three engineers.
- Reduced CS call drop rate in the 2G network by **0.2%**.
- Increased Inter-RAT handover success rate in the UMTS network by **0.8%**, improving overall network performance.

PROJECTS

Object Detection in Fine Art Paintings (MSc Thesis) | *Python, Deep Learning, CNN*

2017 – 2018

- Developed a custom CNN-based approach for object detection in fine art paintings, improving detection accuracy by **18%** over existing methods.
- Built the training and evaluation workflow and performed systematic error analysis to iterate on model design.

EDUCATION

University of Paderborn

Paderborn, DE

Master of Science in Electrical Systems Engineering (Signal Processing)

2015 – 2018

- **Thesis:** Enhancing Object Detection in Fine Art Paintings Using Deep Learning Techniques (+**18%** detection accuracy vs. baseline).
- **Research Assistant:** statistical image processing; plasmonic nanoparticles modeling in COMSOL; Lie groups programming in GAP.

PUBLICATIONS & CERTIFICATIONS

Deep Learning for Object Detection in Fine Art Paintings | *IEEE Xplore*

2020

Nonlinear Dielectric Properties of Random Paraelectric-Dielectric Composites | *Acta Materialia*

2021

Dynamic Retries: failed payment recovery | *Technical writing*

AWS Certified Cloud Practitioner

LANGUAGES

Russian (Native) | English (Fluent) | German (Upper Intermediate, B2)

ADDITIONAL INFORMATION

Volunteer: Bashkortostan branch of the Russian Union of Young Scientists.

Awards: Appreciation letter from the CEO of the Republic of Bashkortostan branch of PJSC “MegaFon” for outstanding performance.